

the upper hand by having way more parallel processing throughput, when we compare the AMD 6990 to the NVIDIA 590 we can see that :

- ▶ AMD Radeon HD 6990: 3072 ALUs x 830 MHz = 2550 billion 32-bit instruction per second
- ▶ Nvidia GTX 590: 1024 ALUs x 1214 MHz = 1243 billion 32-bit instruction per second

Right off these numbers we can see that AMD is ahead by twice the margin. Another factor is how the instruction sets are implemented in the two competing devices. AMD's instruction set is capable doing what NVIDIA's instruction set does in $\frac{1}{3}$ number of steps. This significant advantage in terms of time combined with the sheer raw output puts AMD ahead of NVIDIA.

FPGA

A Field Programmable Gate Array is a chip that comes as a blank slate. You can mount the chip and then program it to do whatever you want to get done. This is the reason why the "Field Programmable" part of the name exists. Now these are fabricated according to order as the number of logic gates on each chip can be customised. Also there are a lot of FPGA packages available for cheap, you can simply purchase them in bulk and start programming each chip to mine bitcoins. They are also quite cheap but since the design isn't streamlined, it isn't as fast as the ASIC but it is way faster than a CPU and comparable to a GPU.

We can see that the Mini Rig by Butterfly Labs is as good as an ASIC miner even though it is FPGA. And most of the FPGA are comparable to GPUs.

ASIC

Application-Specific Integrated Circuit" are basically designed from the ground up to do just one particular task. They are absolutely useless

FPGA	
Avnet Spartan-6 LX150T Development Kit	100
Bitcoin Dominator X5000	100
BitForce SHA256 Single	832
Butterflylabs Mini Rig	25,200
Digilent Nexys 2 500K	5
Icarus	380
KnCMiner Mars	6,000
Lancelot	400
ModMiner Quad	800
Terasic DE2-115	80
X6500 FPGA Miner	400
ZTEX USB-FPGA Module 1.15b	90
ZTEX USB-FPGA Module 1.15x	215
ZTEX USB-FPGA Module 1.15y	860